1641357

22PHYS12

VTU.REG.NO

B. E. First Semester Examination – May 2023

Physics for CSE Stream

(Common for CSE, CSD, ISE and AIML)

Time: 3 hrs] [Maximum Marks: 100

Note: Answer any FIVE full questions, selecting atleast ONE full question from each module.

Physical constants : Velocity of light, $c = 3 \times 10^8$ ms^{-1}; Planck constant, $h = 6.63 \times 10^{-34}$ Js; Boltzman constant , $k = 1.38 \times 10^{-23}$ J/K; Avagadro's number, $N_A = 6.023 \times 10^{26}$ gm/Kmole ; Permittivity of vacuum, $\varepsilon_0 = 8.85 \times 10^{-12}$ Fm^{-1}; Mass of Neutron = 1.67×10^{-27} kg; Acceleration due to gravity , $g = 9.8$ ms^2.

Module - I

1. a) Using neat sketches describe the concstruction and working of GaAs semiconductor diode laser. **(10 Marks)**
   b) Illustrate with neat figure induced absorbtion and stimulated emission in laser. **(06 Marks)**
   c) Find the ratio of population of two energy level in a medium in thermal equilibrium if the transition between them produces light of wavelength 694.3 nm. Assume the ambient temperature as 27°C. $\frac{N_2}{N_1} = 8.87 \times 10^{-31}$ **(04 Marks)**

2. a) Explain the construction of an optical fibre with neat figure and justify why they are called as "Optical waveguides" **(08 Marks)**
   b) With neat figure of an optical fibre arrive at the condition for propagation of a light signal. **(07 Marks)**
   c) An optical fibre is made with core glass if R.I 1.5 and the cladding is doped to give a fractional index difference of 0.0005. Find
   i) R.I of cladding ii) The critical; reflection angle iii) The acceptance angle and iv) The numerical aperture. $n_2 = 1.49$, $\theta_c = 88.2$ N.A = 0.047 $\theta_0 = 2.7°$ **(05 Marks)**

Module - II

3. a) Show that on the basis of uncertiantity principle, a free electron cannot exist within the nucleus of an atom. **(08 Marks)**
   b) Set up one-dimensional time-independent Schrodinger's wave equation. **(07 Marks)**
   c) Calculate the de-Brogile wavelength of the following:
   i) A 1000 kg automobile travelling at 150 ms^{-1}. $\lambda = \frac{h}{mv} = 4.42 \times 10^{-39}$ m
   ii) A 15 g bullet travelling at 500 ms^{-1}. $\lambda = 8.84 \times 10^{-35}$ m
   iii) A dust particle of mass 10^{-6} g moving at 1.5 cms^{-1} $\lambda = 4.42 \times 10^{23}$
   iv) An electron with k,E 1 MeV. $\lambda = 1.228 \times 10^{-12}$ **(05 Marks)**

4. a) Assuming the time independent Schradinger wave equation discuss the solution for energy of a particle in one-dimensional infinite potentional well. **(10 Marks)**
   b) State de-Broglie hypothesis and the conceptual reasoning on which it is based. **(07 Marks)**
   c) Calculate the zero point energy for an electron in a bar of width 10 Å. **(03 Marks)**

$E = \frac{n^2h^2}{8ma^2} = 6.038 \times 10^{-20}$ J

$E_1 = 0.377$ eV

22PHYS

Module - III

5. a) Describe in detail any three polarization mechanisms in dielectric materials. (09 Marks)
   b) Derive Claussius – Mossotti equation. (07 Marks)
   c) The electronic polarization of Krypton gas is 3.54×10^{-40} Fm^2. If the gas contains 2.7×10^{25}/m^3 at NTP, calculate its dielectric constant. (04 Marks)

6. a) Explain the hysteresis property and temperature dependent of dielectric consnat of ferroelectric materials. (10 Marks)
   b) Derive an expression for the internal field in the case of dielectric liquids and solids. (10 Marks)

Module - IV

7. a) Explain the phenomenon of Superconductivity with suitable graph and demonstrate the Meissner effect. (10 Marks)
   b) Discuss any two failures of classical free electron theory. (06 Marks)
   c) In an experiment RED LED (λ = 660mm) is sued for which knee voltage was recorded to be 1.65 V. Calculate Hanck's constant. (04 Marks)

8. a) Derive the expression for number of allowed energy states for electrons in a metal of unit volume. (10 Marks)
   b) Using a semiconductor diode laser and grating (N=2500 LPI) diffraction is produced on a screen placed a distance of 150 cm form grating. If the average distance between central bright spot and 3rd spot is 30 cm. Calculate the wavelength of given laser diode. (10 Marks)

Module - V

9. a) Explain the principles of animation. (10 Marks)
   b) Calculate the Fermi energy of copper wire for the given data below with constant 5×10^{-13}.

Temperature (K)	Voltage (mV)	Current (mA)
303	20.9	47
308	21.2	47
313	21.5	46.9
318	21.9	46.9
323	22.2	46.9
328	22.5	46.9

(10 Marks)

10. a) Explain the modeling of proton decay using Poisson;s distribution. (08 Marks)
    b) In charging and discharging of capacitor experiments 'T_p' was found to be 7s, 15s and 27.5s for C_1, C_2 and C_3 capacitors respectively when R = 100 kΩ. Using the following data determine the dielectric constants of C_1, C_2 and C_3:

Capacitor	C_1	C_2	C_3
Length (mm)	47	114	183
Breadth (mm)	5	5	6
Separation (mm)	0.075	0.075	0.075

(12 Marks)

1641357

22PHYS12

VTU.REG.NO

B.E. First Semester Examination – June 2023(Supplementary)

Physics for CSE Stream

(Common to CSE, CSD, ISE and AIML)

Time: 3 hrs] [Maximum Marks: 100

Note: Answer any FIVE full questions, selecting atleast ONE full question from each module.

Physical constants:

Plank's constant (h) = 6.626070×10^{-34} Js: Charge of electron(e) = 1.6×10^{-19}C;
mass of electron (m) = 9.1×10^{-31}kg, Avogadro number $N_A = 6.022 \times 10^{23}$/mol,
Boltzmann constant (k) = 1.38×10^{-23}J/K, Velocity of light© = 3×10^{8}ms^{-1}

Module – I

1. a) Deduce the expression for energy density of photons in terms of Einstein's A and B co-efficients. **(08 Marks)**
   b) Explain the types of optical fibers with a neat diagram. **(08 Marks)**
   c) The acceptance angle of an optical fiber is 30° in air. Calculate the acceptance angle when the fiber is kept in medium of RI 0.33. **(04 Marks)**

2. a) Explain construction and working of semiconductor diode laser with a neat diagram. **(08 Marks)**
   b) Define the attenuation and hence derive the expression for attenuation. Explain the various causes for optical fiber losses. **(08 Marks)**
   c) Calculate the ratio of population for a given pair of energy levels corresponding to emission of radiation 694.3 nm at a temperature of 300 K. **(04 Marks)**

Module – II

3. a) State and explain Heisenberg's uncertainty principle and hence show that electrons cannot exist in side nucleus. **(08 Marks)**
   b) Derive Eigen values and Eigen functions of a particle inside infinite potential well of infinite height. **(08 Marks)**
   c) An electron is associated with a de Broglie wavelength of 0.1 nm. Calculate the energy and the corresponding momentum of the electron. **(04 Marks)**

4. a) Define phase velocity, group velocity and hence derive the expression de-Broglie's wavelength by analogy. **(08 Marks)**
   b) Derive one-dimensional time independent Schrodinger's wave equation for a matter waves. **(08 Marks)**
   c) Calculate the energy required to rise from ground state to 4th excited state for an electron in one dimensional potential well of width 0.1 nm. **(04 Marks)**

Module – III

5. a) Explain the types of polarization mechanisms. **(08 Marks)**
   b) Explain ferroelectric materials with hysteresis loop. **(08 Marks)**
   c) An elemental dielectric material has polarizability 7×10^{-40} Fm. Assume the internal field to be Lorenz field, calculate the dielectric constant for the material has 3×10^{28} atoms /m^3. **(04 Marks)**

6. a) Derive an expression for Classius-Mossoti equation. **(08 Marks)**
   b) Explain the application of dielectrics in transformers, capacitors, electrical insulation. **(08 Marks)**
   c) A string of identical atoms of spacing 0.2 nm applied with a field of 150 V/m, has polarizability 2.6×10^{-40}Fm. Determine the internal field of this dielectric material. **(04 Marks)**

Module – IV

7. a) Explain the failures of classical free electron theory. **(08 Marks)**
   b) Explain qualitatively BCS theory of superconductor. **(08 Marks)**
   c) Calculate Plank constant of LED, if the knee voltage of red is 1.61 V. (Given wavelength of red is 660 nm) **(04 Marks)**

8. a) Derive an expression for density of states for a solid material of unit volume. (10 Marks)
   b) Calculate the wavelength of laser beam, if the distance between grating element and screen is 150 cm and the first order and second order diffraction maxima is 10 cm and 20 cm. [Given diffraction grating having 2500 lines per inch]. (10 Marks)

Module – V

9. a) Explain with examples of character animation walking and jumping. (05 Marks)
   b) Explain the importance of size and scale and weight and strength in animations. (05 Marks)
   c) Determine the dielectric constant of capacitors using RC circuit. If the length and breadth of capacitor is 47 mm and 5 mm and separation between them is 0.075 mm. Given R = 100 kΩ.

Time in seconds	Charging	Discharging
0	0	10.3
10	6.1	4
20	8.6	1.6
30	9.6	0.6
40	10	0.2
50	10.2	0.1
60	10.3	0.0

(10 Marks)

10. a) Calculate the magnetic field induction by tangent law and experimentally on the axis of coil for a current flowing across the coil is 0.5 A and 1A, if the mean deflection in magnetodeflectometer is 22 and 43°. [Given number of turns = 5, radius of coil = 9.8 cm]. (10 Marks)
    b) Estimate Pi using Monte Corle method. (06 Marks)
    c) Explain descriptive statistics. (04 Marks)